# A Novel Time-varying Continuous Input Shaper for nonlinear systems

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This supplementary file contains complexity analysis, proofs, and several figures and tables related to the experimental setup and results.

## I. Additional Proofs

### S1. Convergence Analysis of HIA-based Parameter Identification

**Assumption 1.** *The search space $\Omega \subset \mathbb{R}^d$ is compact (bounded and closed) and equals the closure of a nonempty bounded open set, so that $\mathrm{int}(\Omega) \neq \emptyset$ and $\mu(\Omega) > 0$.*

**Assumption 2.** *The fitness function $f(\cdot)$ is Lipschitz continuous on $\Omega$, i.e., there exists $L > 0$ such that $|f(x) - f(y)| \leq L\|x - y\|$, $\forall x, y \in \Omega$.*

**Assumption 3.** *Elitist archive: An elitist archive stores the best-so-far individual $T^g$, so that the best fitness value is non-increasing, $f(T^{g+1}) \leq f(T^g)$. Projection and reinitialization: Every candidate generated by Eqs. (11) and (13) is projected onto $\Omega$ before evaluation; any out-of-bound component is reinitialized uniformly at random in $\Omega$ with probability $p_r > 0$, independently of the current state.*

**Theorem 1.** *Under Assumptions 1–3, the elite HIA sequence $\{T^g\}$ converges to the global optimum in probability:*

$$\lim_{g\to\infty} \mathbb{P}\big(|f(T^g) - f^*| < \epsilon\big) = 1, \qquad \forall \epsilon > 0. \tag{S1}$$

*Proof.* Since $\Omega$ is compact and $f$ is continuous, the global minimum $f^* = f(x^*)$ is attained at some $x^* \in \Omega$, and $f\big(T^g\big) \geq f^*$ for all $g$. By the Lipschitz condition, $f(x) < f^* + \varepsilon$ for every $x \in \Omega$ with $\|x - x^*\| < \varepsilon/L$; hence the $\epsilon$-optimal region:

$$S_\epsilon = \{x \in \Omega : f(x) < f^* + \epsilon\}, \tag{S2}$$

contains the set $B(x^*, \epsilon/L) \cap \mathrm{int}(\Omega)$. Since $\mathrm{int}(\Omega) \neq \emptyset$ (Assumption 1) and $x^* \in \Omega = \overline{\mathrm{int}(\Omega)}$, this intersection contains a nonempty open subset of $\mathrm{int}(\Omega)$ (relative topology), so $\mu(S_\epsilon) \geq \mu\big(B(x^*, \epsilon/L) \cap \mathrm{int}(\Omega)\big) > 0$. Defining $c_\epsilon := \mu(S_\epsilon)/\mu(\Omega) > 0$ (with $\mu(\Omega) > 0$ by Assumption 1), the rest follows.

By Assumption 3, in each generation and conditionally on the current population, each of the $N$ candidate slots is a uniform sample on $\Omega$ with probability at least $p_r$, independently across slots. Hence, conditionally on the current state, the probability that no candidate enters $S_\epsilon$ in one generation is at most $(1 - p_r c_\epsilon)^N$, i.e., the per-generation success probability is at least

$$q_\epsilon := 1 - (1 - p_r c_\epsilon)^N > 0. \tag{S3}$$

By the elitist archive (Assumption 3), once a candidate with $f < f^* + \varepsilon$ has been evaluated, $f\big(T^h\big) < f^* + \varepsilon$ for all $h$ thereafter. Since $q_\epsilon > 0$ is uniform over states, a standard conditional Markov bound gives:

$$\mathbb{P}\big(|f(T^g) - f^*| \geq \epsilon\big) \leq (1 - q_\epsilon)^g \xrightarrow[g\to\infty]{} 0, \tag{S4}$$

which yields the claim. Note that the argument relies only on the elitist archive and the positive-probability uniform reinitialization (Assumption 3); the DE/BAS operators and the adaptive $F_j, CR_j$ only accelerate the search and do not affect the convergence argument. □ □

### S2. Computational Stability of the Proposed Update Rule in Eqs. (9)–(14)

**Assumption 4.** *Let $M := \max_{x\in\Omega} \|x\| < \infty$.*

**Assumption 5.** *$F_j \in [F_{\min}, F_{\max}]$ with $F_{\max} \leq 0.9$ (here $[F_{\min}, F_{\max}] = [0.3, 0.9]$), and $CR_j \in [0, 1]$.*

**Assumption 6.** *The BAS parameters are finite: antennae distance $d_0 > 0$ and step size $s \leq s_0 < \infty$ at all generations.*

**Theorem 2.** *Under Assumptions 1–6, all quantities computed by Eqs. (9)–(14) are, at every generation $g$, uniformly bounded by the generation-independent constant*

$$\overline{M} := \max\Big\{\sqrt{(1 + 2F_{\max})^2 + 1}\, M,\ M + d_0,\ M + s_0,\ \|f\|_\infty,\ 1\Big\} < \infty, \tag{S5}$$

*where* $\|f\|_\infty := \max_{x\in\Omega}|f(x)| < \infty$ *(by continuity of* $f$ *on compact* $\Omega$*). Hence Eq. (11) has stable computational behavior and no computational overflow can occur.*

*Proof. (i) Population.* By Assumption 3 every candidate is projected onto $\Omega$ before evaluation, and the selection rule of Eq. (14) only accepts projected candidates. An induction over $g$ (with initialization in $\Omega$ by Assumption 4) gives $\|T_i^g\|_2 \le M$ for all $i$ and all $g$.

*(ii) Mutant vector.* For $v_j = T_a + F_j(T_b - T_c)$ in Eq. (11), the triangle inequality and (i) give

$$\|v_j\|_2 \le \|T_a\|_2 + F_j\|T_b - T_c\|_2 \le M + F_j(\|T_b\|_2 + \|T_c\|_2) \le (1+2F_j)M \le (1+2F_{\max})M, \tag{S6}$$

where $F_j \le F_{\max} \le 0.9$ by Assumption 5.

*(iii) Trial vector.* Each component of $u_j$ in Eq. (11) satisfies $u_{j,d} \in \{v_{j,d}, T_{j,d}\}$, hence $u_{j,d}^2 \le \max\{v_{j,d}^2, T_{j,d}^2\} \le v_{j,d}^2 + T_{j,d}^2$. Summing over $d$,

$$\|u_j\|_2^2 = \|v_j\|_2^2 + \|T_j\|_2^2 \le \big((1+2F_{\max})^2 + 1\big)M^2, \tag{S7}$$

so $\|u_j\|_2 \le \sqrt{(1+2F_{\max})^2+1}\,M$. With $F_{\max} = 0.9$ this is $\approx 2.973M$.

*(iv) BAS quantities.* By Eqs. (12)–(13), $\|d\|_2 = 1$, and Assumption 6 ($d_0 > 0$, $s \le s_0 < \infty$),

$$\|T_j^{\text{left}}\|_2,\ \|T_j^{\text{right}}\|_2 \le M + d_0, \qquad \|T_j^B\|_2 \le M + s_0. \tag{S8}$$

*(v) Adaptive parameters.* Let $f_{\max}^{\text{pop}} := \max_j f(T_j)$ and $f_{\min}^{\text{pop}} := \min_j f(T_j)$ denote the current-population extrema (as in Eq. (9)), which are finite since $f$ is continuous on compact $\Omega$ and $T_j \in \Omega$. Then $0 \le f_{\max}^{\text{pop}} - f_{\min}^{\text{pop}} < \infty$; with $\kappa > 0$ the denominator of Eq. (9) never vanishes and

$$0 \le \lambda_j = \frac{f(T_j) - f_{\min}^{\text{pop}}}{f_{\max}^{\text{pop}} - f_{\min}^{\text{pop}} + \kappa} \le \frac{f_{\max}^{\text{pop}} - f_{\min}^{\text{pop}}}{f_{\max}^{\text{pop}} - f_{\min}^{\text{pop}} + \kappa} < 1. \tag{S9}$$

Hence $F_j \in [F_{\min}, F_{\max}]$ and $CR_j \in [0,1]$ by Eq. (10).

Combining (i)–(v), every quantity produced by Eqs. (9)–(14) is bounded by $\overline{M}$ at every generation; no unbounded growth or floating-point overflow can occur. The elitist archive (Assumption 3) additionally keeps the best fitness non-increasing, $f(T^{g+1}) \le f(T^g)$. □

□ □

## II. Complexity analysis

The initialization and initial evaluation require $\mathcal{O}(N \cdot d + N \cdot |\phi|)$, where $N$ is the population size, $d$ is the individual dimension ($d = 6$ for parameter identification and $d = m$ for delay optimization), and $|\phi|$ is the number of training samples, so that one evaluation of the vibration function $\mathbf{H}(\cdot)$ costs $\mathcal{O}(|\phi|)$. In each generation, DE mutation/crossover costs $\mathcal{O}(N \cdot d)$, and the fitness evaluations, i.e., one trial vector plus two BAS antennae per individual, cost $\mathcal{O}(N \cdot |\phi|)$ (a constant factor of about 3 is absorbed). Hence each generation costs $\mathcal{O}(N \cdot (d + |\phi|))$. Assuming at most $n$ iterations, the overall time complexity is given as:

$$\mathcal{O}(N \cdot d + N \cdot |\phi|) + \mathcal{O}\big(n \cdot N \cdot (d + |\phi|)\big) = \mathcal{O}\big(n \cdot N \cdot (d + |\phi|)\big) = \mathcal{O}(n \cdot N \cdot |\phi|), \tag{S10}$$

since $d$ is constant and $n \cdot N \cdot d$ is dominated by $n \cdot N \cdot |\phi|$.

## III. Additional Tables

Table S.I compares the proposed method with existing input shaping approaches. Unlike ZV/ZVD/EI/UMZV shapers, TVCIS uses continuous time-varying shaping functions and is effective in suppressing multi-frequency vibrations. Unlike existing CIS, TVCIS introduces tunable delays for each mode and a modal weighting factor, enabling adaptation to time-varying dynamics. Unlike GA/PSO/DE/BAS-based identification, the proposed HIA integrates DE's mutation and crossover with BAS's directional local search. This hybrid structure is novel for joint optimization of shaping delays and system parameters.

TABLE S.I
COMPARISON OF REPRESENTATIVE NONLINEAR VIBRATION CONTROL STRATEGIES AND THE PROPOSED TVCIS

| Reference | Method | Advantages | Disadvantages | Relation to this work |
|---|---|---|---|---|
| **[1]** | Dynamic bistable electromagnetic energy harvester for ultra-low-frequency rotation | Dynamic bistability lowers resonant frequency; high output power below 1 Hz | Harvesting-oriented; passive structure without suppression adaptivity | Nonlinear bistability aids broadband energy capture; TVCIS attains adaptivity via parameter identification |
| **[2]** | Two-stage hybrid vibration isolator with NES and negative-stiffness spring | Broadband low-frequency isolation; lower force transmissibility | Theoretical model only; chaos/bifurcation complicates tuning | Passive nonlinear isolation; TVCIS shapes the command signal feedforwardly |
| **[3]** | Bionic magnetoelastic bi-stable absorber with tunable stiffness | Broadband targeted energy transfer at small mass ratio; experimentally validated | Extra magnetic hardware; tuning depends on measured excitation | Adjustable stiffness parallels TVCIS adaptivity, but TVCIS is sensor-free feedforward |
| **[4]** | Piecewise quad-stable piezoelectric energy harvester | Broadband energy conversion; baffles suppress large resonance | Complex parameter design; limited low-frequency output | Shares the broadband idea; TVCIS suppresses rather than harvests vibration |
| **[5]** | Multi-stiffness combination absorber with energy harvesting | Synergistic suppression and harvesting; shock robust; active modulation | Hardware complexity; needs excitation monitoring | Active-passive integration idea related; TVCIS adapts without added absorber hardware |
| **[6]** | Review of energy-shunt-inspired vibration suppression | Unified energy-based framework; design guidance for nonlinear DVAs | Passive absorbers rely on frequency tuning; poor adaptivity to time-varying excitation | Motivates the adaptive time-varying feedforward shaping of TVCIS |

TABLE S.II
COMPARISON OF THE PROPOSED TVCIS WITH EXISTING METHODS

| Method | Continuous | Time-varying | Multi-frequency | Adaptive parameters | Parameter identification | Hybrid method |
|---|---|---|---|---|---|---|
| **ZV/ZVD/EI/UMZV** | – | – | – | – | – | – |
| **CIS** | ✓ | – | – | – | – | – |
| **GA/PSO/DE/BAS-based IS** | – | Partial | – | ✓ | ✓ | – |
| **Proposed TVCIS** | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

TABLE S.III
ESTIMATED ACCURACY (IN G) OF METHODS M6-M11 ON MAE, RMAE AND MAX

| NO. | Metric | Baseline | PSO | GA | DE | BAS | HIA |
|---|---|---|---|---|---|---|---|
| **D1** | MAE | 1.78±0.09 | 0.25±0.02 | 0.33±0.04 | 0.22±0.03 | 0.25±0.03 | •0.19±0.03 |
| | RMSE | 1.72±0.01 | 0.31±0.02 | 0.36±0.05 | 0.28±0.03 | 0.29±0.04 | •0.24±0.03 |
| | MAX | 2.88±0.02 | 0.62±0.05 | 0.84±0.09 | 0.59±0.08 | 0.60±0.05 | •0.54±0.07 |
| **D2** | MAE | 0.96±0.00 | 0.06±0.01 | 0.04±0.07 | 0.04±0.02 | 0.04±0.00 | •0.03±0.00 |
| | RMSE | 0.96±0.00 | 0.07±0.01 | 0.06±0.07 | 0.05±0.01 | 0.05±0.01 | •0.04±0.01 |
| | MAX | 1.95±0.03 | 0.15±0.05 | 0.15±0.10 | 0.12±0.05 | 0.12±0.02 | •0.11±0.00 |
| **D3** | MAE | 0.31±0.00 | 0.04±0.03 | 0.05±0.05 | 0.04±0.02 | 0.05±0.01 | •0.03±0.03 |
| | RMSE | 0.33±0.00 | 0.05±0.02 | 0.06±0.04 | 0.05±0.01 | 0.05±0.00 | •0.04±0.01 |
| | MAX | 1.15±0.00 | 0.11±0.07 | 0.13±0.09 | 0.09±0.02 | 0.12±0.01 | •0.08±0.02 |
| **D4** | MAE | 1.26±0.02 | 0.10±0.01 | 0.10±0.02 | 0.09±0.01 | 0.10±0.03 | •0.08±0.05 |
| | RMSE | 1.28±0.02 | 0.10±0.01 | 0.10±0.01 | 0.10±0.00 | 0.10±0.01 | •0.08±0.04 |
| | MAX | 1.65±0.01 | 0.35±0.02 | 0.36±0.09 | 0.33±0.05 | 0.35±0.05 | •0.31±0.09 |
| **R+** | | 78 | 78 | 78 | 78 | 78 | - |
| **R−** | | 1 | 1 | 1 | 1 | 1 | - |
| ***p*-value** | | 2.44E−4 | 2.44E−4 | 2.44E−4 | 2.44E−4 | 2.44E−4 | - |

Note that Baseline represents the performance of the model before system parameter identification. The Wilcoxon signed-rank test is conducted at a significance level of 0.05. Besides, • denotes the best performance value for each dataset.

## IV. Additional Figures

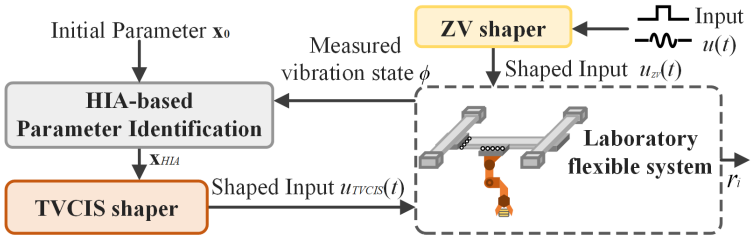
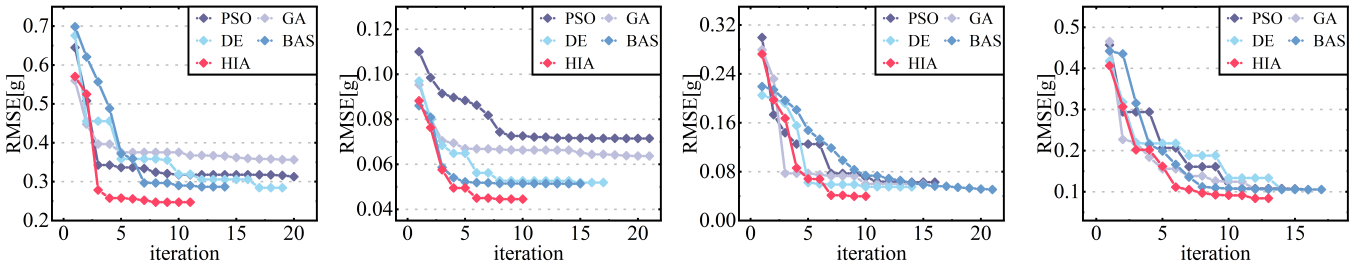
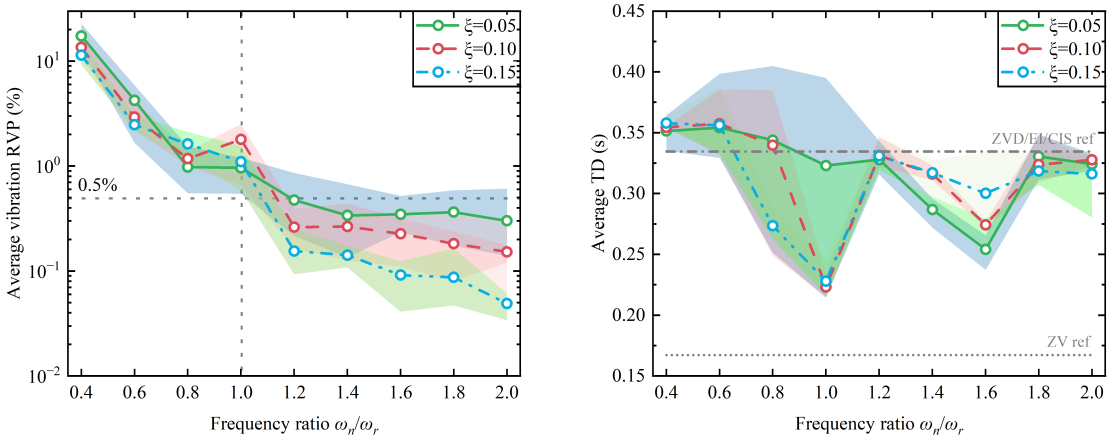



Fig. S.1. Workflow chart for TVCIS model.



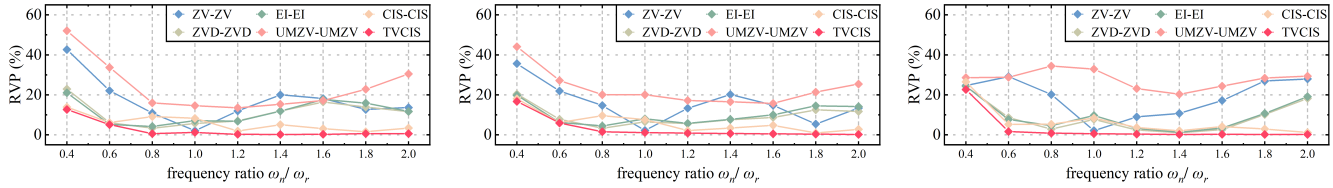

(a) RMSE on D1 (b) RMSE on D2 (c) RMSE on D3 (d) RMSE on D4

Fig. S.2. Comparison of the training curves of HIA and its comparators across D1–D4 in terms of RMSE. HIA exhibits the fastest convergence because the adaptive $F$ and $CR$ mechanism assigns larger $F$ and smaller $CR$ to individuals far from the best solution (global exploration) and smaller $F$ and larger $CR$ to those close to it (local exploitation), while the DE mutation/crossover is combined with the BAS directional local search and the elitist selection in Eq. (14) retains the better candidate. From the control perspective, this means that the identified natural frequencies and damping ratios become accurate enough for the shaper to remain effective within fewer iterations, which is important when the shaper must be re-tuned as the operating condition changes.

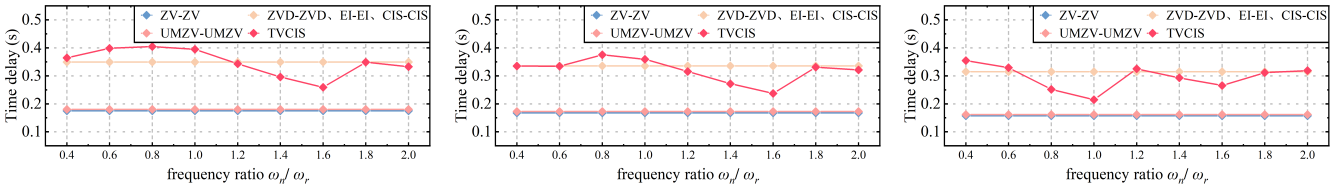



(a) RVP of TVCIS vs. frequency ratio (b) Time delay of TVCIS vs. frequency ratio

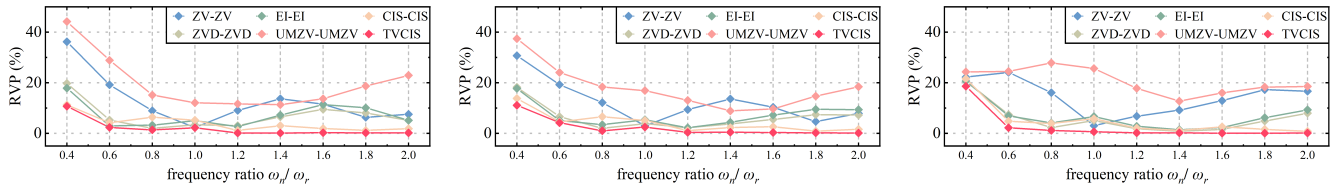
Fig. S.3. Sensitivity of TVCIS to system parameters. (a) RVP and (b) TD versus frequency ratio $\omega_n/\omega_r$ for $\xi$ = 0.05, 0.10 and 0.15. Markers are means over the three configurations $(\omega_n^1, \omega_n^2) = (30, 45)$, $(30, 50)$, and $(30, 60)$ Hz and bands their min-max range ($n$=3); Dashed lines in (b) give the constant ZV (0.167 s) and ZVD/EI/CIS (0.334 s) delays at $\xi$ = 0.10. This behavior follows from the delay-tuning mechanism: the delay is determined by the requirement that the execution duration covers a full vibration period so that the integral coefficients $a_{0k}$ in Eq. (7) drive the modal free response to zero, and the resulting delay is therefore governed mainly by the modal frequency rather than by the damping ratio.



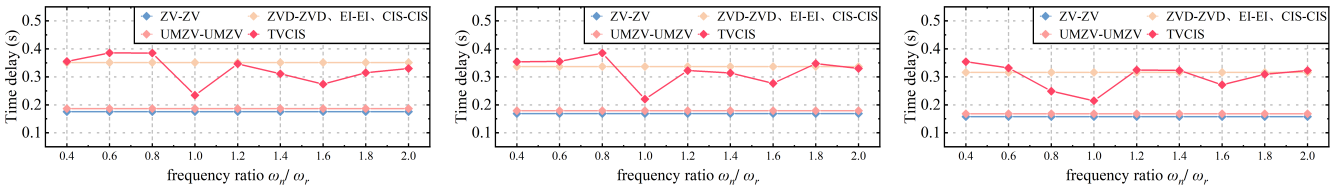

(a) RVP with $(\omega_n^1, \omega_n^2) = (30, 45)$, $\zeta = 0.05$ (b) RVP with $(\omega_n^1, \omega_n^2) = (30, 50)$, $\zeta = 0.05$ (c) RVP with $(\omega_n^1, \omega_n^2) = (30, 60)$, $\zeta = 0.05$



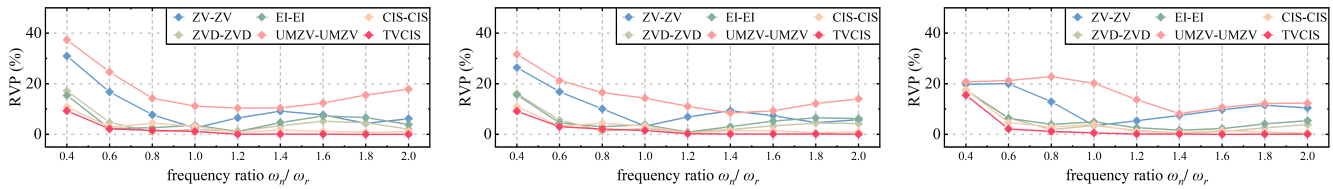

(d) TD with $(\omega_n^1, \omega_n^2) = (30, 45)$, $\zeta = 0.05$ (e) TD with $(\omega_n^1, \omega_n^2) = (30, 50)$, $\zeta = 0.05$ (f) TD with $(\omega_n^1, \omega_n^2) = (30, 60)$, $\zeta = 0.05$



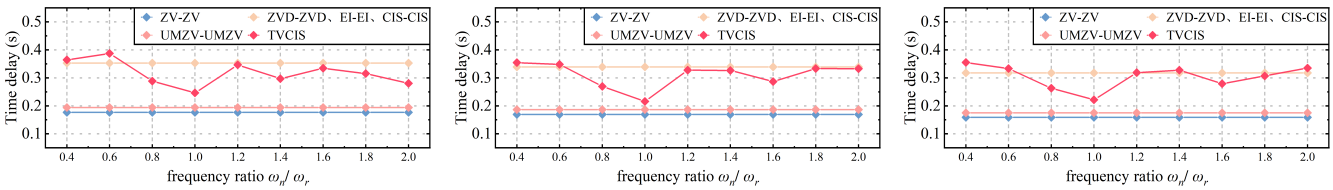

(g) RVP with $(\omega_n^1, \omega_n^2) = (30, 45)$, $\zeta = 0.1$ (h) RVP with $(\omega_n^1, \omega_n^2) = (30, 50)$, $\zeta = 0.1$ (i) RVP with $(\omega_n^1, \omega_n^2) = (30, 60)$, $\zeta = 0.1$




(j) TD with $(\omega_n^1, \omega_n^2) = (30, 45)$, $\zeta = 0.1$ (k) TD with $(\omega_n^1, \omega_n^2) = (30, 50)$, $\zeta = 0.1$ (l) TD with $(\omega_n^1, \omega_n^2) = (30, 60)$, $\zeta = 0.1$




(m) RVP with $(\omega_n^1, \omega_n^2) = (30, 45)$, $\zeta = 0.15$ (n) RVP with $(\omega_n^1, \omega_n^2) = (30, 50)$, $\zeta = 0.15$ (o) RVP with $(\omega_n^1, \omega_n^2) = (30, 60)$, $\zeta = 0.15$




(p) TD with $(\omega_n^1, \omega_n^2) = (30, 45)$, $\zeta = 0.15$ (q) TD with $(\omega_n^1, \omega_n^2) = (30, 50)$, $\zeta = 0.15$ (r) TD with $(\omega_n^1, \omega_n^2) = (30, 60)$, $\zeta = 0.15$

Fig. S.4. The simulation experiment results. TVCIS attains the lowest RVP in every case, most notably at off-tuned ratios where fixed-impulse and static CIS shapers degrade, while its TD remains comparable to that of ZVD/EI/CIS and nearly invariant with respect to $\omega_n/\omega_r$, confirming a superior RVP–TD trade-off and robustness to parameter mismatch. This is attributed to the continuous, time-varying nature of the shaping function: instead of concentrating the command energy at discrete impulse instants as ZV/ZVD/EI/UMZV do, it spreads the command energy away from the structural resonances and provides smooth signal transitions, thereby suppressing the residual vibration even when the actual frequency deviates from the design value. The comparison also shows why the negative-impulse UMZV and the classical ZV shaper degrade under frequency detuning and multi-modal coupling: their fixed impulse structures are designed for a single dominant frequency and cannot redistribute energy across modes. This clarifies the control-relevant advantage of the proposed scheme.